

Anorthite 743: crystal morphology

Groth/Krantz SHAPE library model 743 (743.SHP), anorthite, drawn from its SHAPE input as supplied

Anorthite model 743 from the Groth/Krantz crystal library for SHAPE. It has 21 forms (42 faces) and is the geometry used by the **anorthite_743** pymsh class, whose corners are identical to the SHAPE orientated listing (An_743_orientated.txt). The crystal is shown exactly as the SHAPE file defines it, on the 14.16 Å cell given in the file.

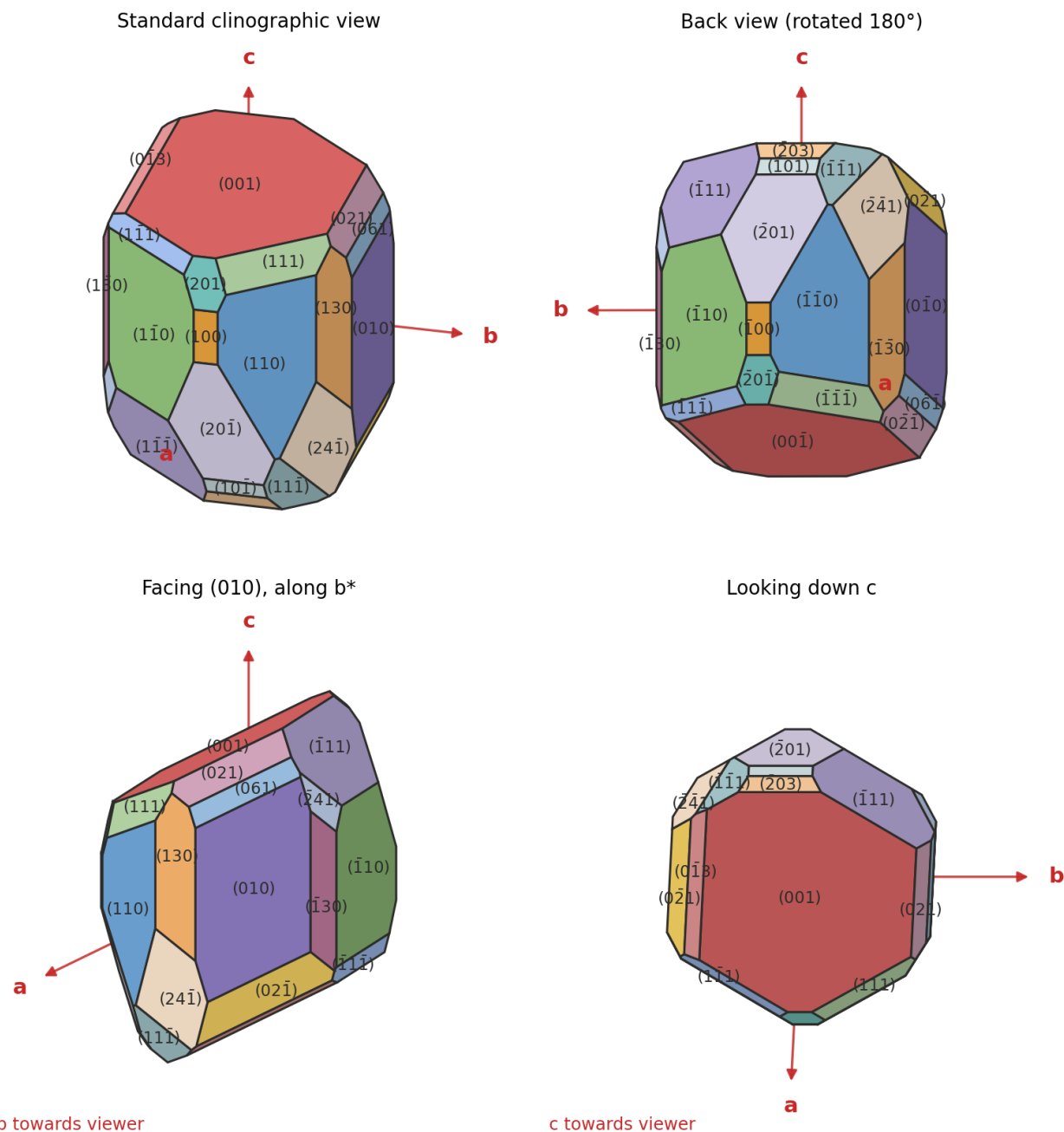


Figure 1. Orthographic views of the crystal in the frame used by SHAPE and the pymsh classes (c along z, b in the yz plane, centre at the origin). Red arrows are the crystal axes; faces with the same colour belong to the same form.

Crystal data

Quantity	Value
SHAPE file	743.SHP (title 743Ano, class code 2B1)
Unit cell a, b, c (Å)	8.18, 12.88, 14.16
α , β , γ (°)	93.15, 115.9, 91.2
Point group / morphology symmetry	-1 (triclinic, centrosymmetric)
Faces, corners, edges	42, 80, 120
Angle between c and the (001) normal	26.20°
pygmsh class	anorthite_743 (xtal_geometries2.py)

Forms

Form	Central distance	Faces	Share of surface
{0 0 1}	0.48000	2	26.1%
{0 1 0}	0.50000	2	12.3%
{-1 1 0}	0.50000	2	11.3%
{1 1 0}	0.52000	2	8.1%
{-1 1 1}	0.59000	2	7.3%
{-2 0 1}	0.57000	2	5.7%
{-2 -4 1}	0.57000	2	3.9%
{1 3 0}	0.53500	2	3.9%
{0 -2 1}	0.61000	2	3.3%
{-1 3 0}	0.55000	2	2.6%
{1 1 1}	0.55000	2	2.5%
{0 2 1}	0.58500	2	2.4%
{0 6 1}	0.53500	2	1.6%
{0 -1 3}	0.58500	2	1.5%
{1 -1 1}	0.56200	2	1.5%
{-1 -1 1}	0.69000	2	1.5%
{2 0 1}	0.56500	2	1.1%
{-2 0 3}	0.74000	2	0.9%
{1 0 0}	0.56000	2	0.8%
{-2 4 1}	0.57500	2	0.8%
{-1 0 1}	0.68000	2	0.7%

Central distances are those in the SHAPE file (arbitrary units). Surface shares are computed from the polyhedron built from these forms, which reproduces SHAPE's own corners to 10^{-7} .

Size and shape when the pygmsh class is called with $z = 1$

Quantity	Value
Scaling	$sf = z / f_{sp_z}$ with $f_{sp_z} = 0.704836$ (half-length along c in SHAPE units)
Length along c	2.000 (spans -1 to +1 along z)
Width (perpendicular to c, in (010))	1.592
Thickness (perpendicular to (010))	1.419
Length : width : thickness	1.41 : 1.12 : 1
Half-extents $ x $, $ y $, $ z $	0.795, 0.725, 1.000
Volume V1	2.3943
Surface area	9.5484
Equivalent-sphere radius	0.8299
Surface / volume	3.988

To match a measured length L along c use $z = L/2$; to match a measured volume V use $z = (V / V1)^{1/3}$.

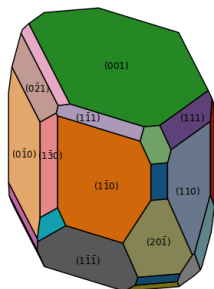
Source and notes

- Geometry: built from the 21 forms and central distances in 743.SHP. The resulting 80 corners match SHAPE V7.3 output (An_743_orientated.txt) to 10^{-7} .
- Scaling in the pygmsh class: fsp_z = 0.704836, the half-length along c (previously 0.48, the (001) distance).
- **Caveat on indexing.** The cell in 743.SHP has $c = 14.16 \text{ \AA}$, but the face symbols (for example $(-2\ 0\ 1)$, (021) , $(-2\ 4\ 1)$) are the classic morphological symbols of anorthite, which refer to axes with $c \approx 7.08 \text{ \AA}$ (axial ratio $a:b:c = 0.6347 : 1 : 0.5501$, 1911 Encyclopaedia Britannica). If the symbols were intended for the $7 \text{ \AA}$ axes, SHAPE builds a somewhat different crystal from the one Groth modelled. The final page compares the two readings.
- The model is idealised (a teaching model, not a measured crystal).

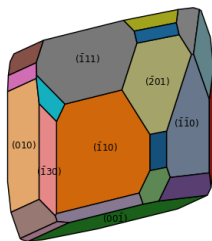
Effect of the c-axis length used with the face symbols

Groth anorthite 743: effect of the c-axis length used with its face indices (c axis vertical, orthographic)

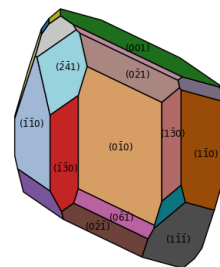
743.SHP as written ($c = 14.16 \text{ \AA}$): front



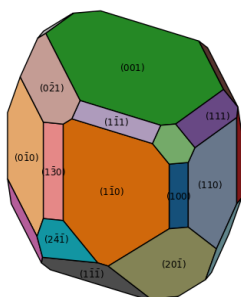
743.SHP as written ($c = 14.16 \text{ \AA}$): back



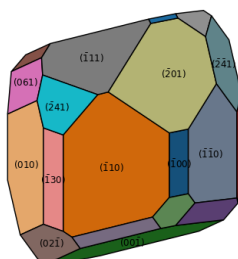
743.SHP as written ($c = 14.16 \text{ \AA}$): facing $(0\bar{1}0)$



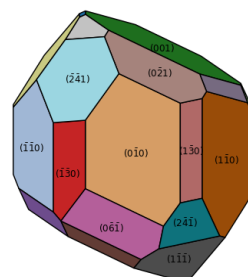
Same faces on the morphological axes ($c = 7.08 \text{ \AA}$): front



Same faces on the morphological axes ($c = 7.08 \text{ \AA}$): back



Same faces on the morphological axes ($c = 7.08 \text{ \AA}$): facing $(0\bar{1}0)$



Top: 743.SHP as written ($c = 14.16 \text{ \AA}$), which is the geometry of the pygmsh class. Bottom: the same face symbols read on the morphological axes ($c = 7.08 \text{ \AA}$).

SHAPE input file (crystal definition)

743Ano								
2B1	0	1	3.10	0				
8.1800000	12.8800000	14.1600000						
93.1500000	115.9000000	91.2000000						
1	0	0	.56000	0	0	0	0	0
1	1	0	.52000	0	0	0	0	0
-1	1	0	.50000	0	0	0	0	0
0	1	0	.50000	0	0	0	0	0
0	0	1	.48000	0	0	0	0	0
2	0	1	.56500	0	0	0	0	0
1	3	0	.53500	0	0	0	0	0
-1	3	0	.55000	0	0	0	0	0
1	1	1	.55000	0	0	0	0	0
1	-1	1	.56200	0	0	0	0	0
0	2	1	.58500	0	0	0	0	0
0	-2	1	.61000	0	0	0	0	0
0	6	1	.53500	0	0	0	0	0
0	-1	3	.58500	0	0	0	0	0
-1	1	1	.59000	0	0	0	0	0
-1	-1	1	.69000	0	0	0	0	0
-2	0	3	.74000	0	0	0	0	0
-2	0	1	.57000	0	0	0	0	0
-2	4	1	.57500	0	0	0	0	0
-2	-4	1	.57000	0	0	0	0	0
-1	0	1	.68000	0	0	0	0	0
99	99	99	1.00000	0	0	0	0	0

Line 1: title. Line 2: crystal class code (2 = triclinic -1) and SHAPE options. Lines 3-4: a, b, c and alpha, beta, gamma. Following lines: h, k, l and central distance of each form, ending with 99 99 99. The display settings that follow in the file are not shown.